AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Midterm Examination (MTA) – Makeup
Spring Semester 2025/2026

الجامعة العربية المفتوحة - مصر
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Arab Republic of Egypt

Date: 28/Apr./2026

umber of Exam Pages: 5 (cluding this cover sheet)	**Time Allowed:** 2 Hours

structions:

- *Total Marks: 60* **marks**
- *This exam consists of* **6 questions**
- ***ALL questions*** *must be answered in the External Answer booklet.*
- *Be sure you write your* **name and ID** *on the External Answer booklet.*
- ***Basic Calculators*** *are allowed.*

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

1) _____ is one of the categories of putting together a business case for accessibility, this category means the opportunities of increased website usage and Search Engine Optimisation, etc.
   A. Social factors.
   B. Technological factors.
   C. Financial factors.
   D. Legal and policy factors.

2) The XHTML Basic document type _____.
   A. includes the maximum set of modules required to be an XHTML client language document type, but it does not include images, forms, basic tables, and object support.
   B. includes the minimal set of DOM modules that are needed to display the tags such as images, forms, basic tables, and required to be an XHTML host language document type.
   C. includes all modules of DOM and XHTML client language required to support forms, basic tables, images, window, document, and navigator objects.
   D. includes the minimal set of modules required to be an XHTML host language document type, and in addition it includes images, forms, basic tables, and object support.

3) _____ is the balance and interplay of letterforms on the page—a verbal and visual equation that helps the reader understand the form and absorb the substance of the page content.
   A. Portability.
   B. Legibility.
   C. Accountability.
   D. Typography.

4) A _____ client performs a significant amount of processing on the client side and thus puts greater demands on the client-side machine.
   A. purely.
   B. thin.
   C. thick.
   D. trusted.

5) The document object _____.
   A. is a child of the navigator object, and allows us to manipulate a web page. For example, you can use the document object to insert content in a web page.
   B. allows us to retrieve information about the browser that has been used to access a page, you can then identify the name of the browser, and the operating system of the client.
   C. is at the top level of the DOM hierarchy is a single window object, which represents the browser window. All the other objects in the DOM are contained directly or indirectly by the window object.
   D. is a child of the window object, and allows us to manipulate a web page. For example, you can use the document object to insert content in a web page.

6) A key is used to _____.
   A. improve efficiency in searching and sorting; only one column can be a key.
   B. improve security during data operations; keys are used to protect records.
   C. improve the efficiency of operations such as searching and sorting, one or more columns can be specified to be a 'key'.
   D. improve speed in searching and sorting; only one column can be a key.

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7) Presentation layer or tier _____.
A. is quite commonly called for the server application, or software, that displays information.
B. is simple client server architecture which is also commonly called 'two-tier architecture' to communicate between server and client.
C. is quite commonly called for the client application, or software, that displays information.
D. is a protocol that handles the case where the client is loosely coupled to the server application.

8) Mobile Web is a solution to _____.
A. providing internet for remote communities. This means creating a pared down version of XHTML that can be readily interpreted by many different devices. To this end the W3C created the XHTML Basic document type.
B. providing internet for local communities. This means creating a complete version of XHTML that can be readily interpreted by specific devices in that local area. To this end the CERN created the XHTML Basic document type.
C. providing internet for local communities. This means creating a complete version of XHTML that can be readily interpreted by many different devices in that area. To this end the CERN created the XHTML Basic document type.
D. providing internet for remote communities. This means creating a complete version of HTML that can be readily interpreted by specific devices in that area. To this end the W3C created the HTML Basic document type.

9) The values from the form fields will be available for a PHP script as elements in the _____ superglobal.
A. attribute.
B. record.
C. post.
D. content.

10) In the following code:
```
if (!$connection) {
  Print ("Internal error". mysql_errno()); }
```
A. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case an error message is printed and the 'mysql_error()' will be called.
B. The code has a conditional expression '!' (i.e. not) connection that will be false if 'connection' evaluates to true. In this case an error message is printed and the 'mysql_error()' will be called.
C. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case, the connection will be established.
D. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case an error message is printed and the 'mysql_error()' will be called then the connection will be established.

PART 2: ALL QUESTIONS ARE REQUIRED [20 Marks]

Question 2: (12 marks 3 marks each one)

a) List three of Neilsen quality components that define usability
b) List three of the distinct layers of Web service architecture (WSA) which is a particular realisation of SOA that is used to provide services over the internet

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c) List three of the guidelines areas covered by the Web Accessibility Initiative.
d) What is the difference between the two functions that are used in PHP for including a file.

Question 3: (8 marks 4 marks each one)

a) In general, the operation of web services involves several processes, discuss four of these general operations of web services.
b) In general, two types of computer languages can be distinguished, discuss these two types of computer languages.

PART 3: ALL QUESTIONS ARE REQUIRED [30 Marks]

Question 4: (5 marks)

Write the appropriate SQL statements to perform the following requirements:

1. Create a table called 'MyCourses' with the following fields using any appropriate data types: CourseName, and MTA.
2. Insert a row into the table 'MyCourses' with the following values. CourseName = "TT284", MTA = 25.5.
3. Display all the data in the table 'MyCourses' for the course 'TT284'.
4. Display CourseName , FinalExam in the table 'MyCourses' in descending order based on MTA field.

Question 5: (15 marks 5 marks each one)

a) Write HTML code combined with CSS code to set attributes of Header 1 element as follows:

1. font type is Algerian.
2. font-size: 40pt;
3. color is green.
4. alignment is center.

and finally type "MTA is today" as header 1 that will have all the above formats.

b) If you have the following HTML code and it was saved as Q5b.HTML:

```
<html>
  <head> <script type = "text/javascript" src="MyFunctions.js"></script>  </head>
  <body>
    <script>checkLength("12",3,4);</script>
    <script>checkLength("1234",2,4);</script>
   </body>   </html>
```

Then we created the Java script file MyFunctions.JS that contains the following function

```
function checkLength(value, min, max) {
 if (value.length >= min && value.length <= max)  {
   document.write("OK <br>");
 }
 else  {
   document.write("Please enter value between " + min + " and " + max + "
characters" + "<br>");
 }}
```


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What will be the output if we run file Q5b.HTML?

c) Write a PHP code to perform the following:
   1. Create an associative array named StudentsNumbers that having the following names of year and associated number of students: "Freshman"=>"1000", "Sophomore"=>"800", "Junior "=>"600","Senior"=>"400".
   2. Calculate the total and the average number of all students.
   3. Print the total and the average number of all students.

Question 6: (10 marks)

Develop a web page containing the following specification:

1. Page Title "Grades Management System"
2. Header 2 "Grades System"
3. Form action to locate the script that will process the form data "http//norton.open.ac.uk/reflect.php"
4. Text Input type field for the user name.
5. Password type field for the password.
6. List type
   for the Tutor Role
   for the Student Role
7. Hidden field to post the value "Accepted".
8. Submit button with Submit name.
9. Reset Button with Reset name.

The web page should look like the following design (Figure 1):

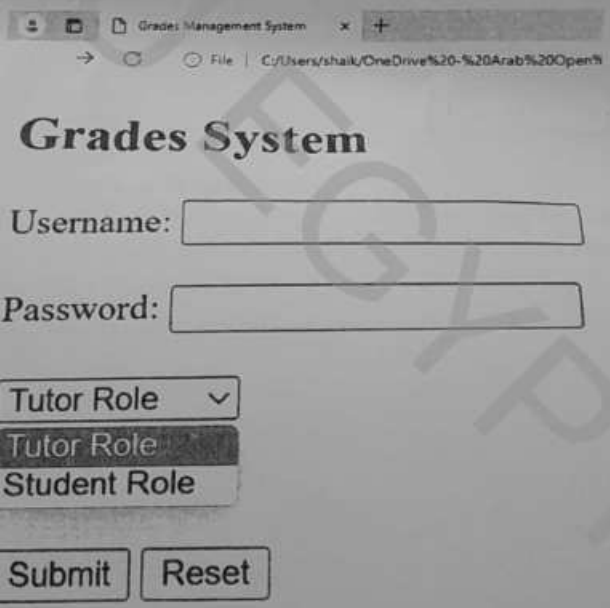


Figure 1

End of the questions.

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